```
# ds experiment 10
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris

# Load dataset
iris = load_iris()

# Convert to DataFrame
df = pd.DataFrame(iris.data, columns=iris.feature_names)

# Add target column
df['species'] = iris.target

# Display first 5 rows
print("First 5 Rows:")
print(df.head())

# Basic information
print("\nDataset Info:")
print(df.info())

print("\nStatistical Summary:")
print(df.describe())
```

```
First 5 Rows:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   species
0        0
1        0
2        0
3        0
4        0

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   sepal length (cm)  150 non-null    float64
 1   sepal width (cm)   150 non-null    float64
 2   petal length (cm)  150 non-null    float64
 3   petal width (cm)   150 non-null    float64
 4   species            150 non-null    int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB
None

Statistical Summary:
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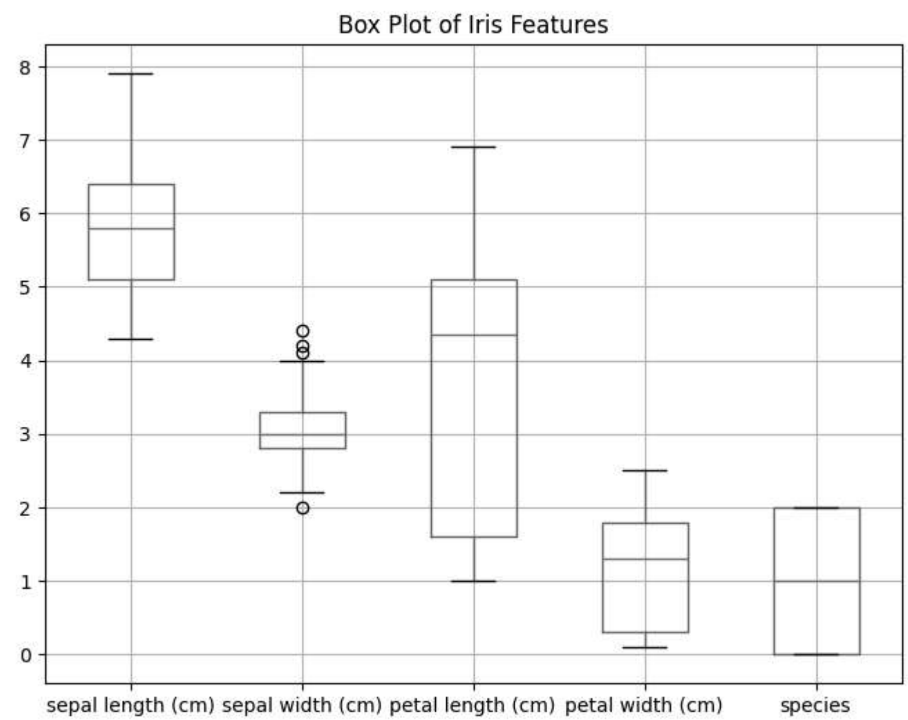
```
       sepal length (cm)  sepal width (cm)  petal length (cm)  \
count         150.000000        150.000000         150.000000   
mean            5.843333          3.057333           3.758000   
std             0.828066          0.435866           1.765298   
min             4.300000          2.000000           1.000000   
25%             5.100000          2.800000           1.600000   
50%             5.800000          3.000000           4.350000   
75%             6.400000          3.300000           5.100000   
max             7.900000          4.400000           6.900000   

       petal width (cm)     species  
count        150.000000  150.000000  
mean           1.199333    1.000000  
std            0.762238    0.819232  
min            0.100000    0.000000  
25%            0.300000    0.000000  
50%            1.300000    1.000000  
75%            1.800000    2.000000  
max            2.500000    2.000000  
```

```
# Box plot
plt.figure(figsize=(8,6))
df.boxplot()
plt.title("Box Plot of Iris Features")
plt.show()
```




```
# Plot histograms
df.hist(figsize=(10,8))
```

```
plt.suptitle("Histograms of Iris Features")
plt.show()
```